

Validation Framework for Synthetic Occupancy Schedules in Building Energy Modeling

The reliability of building energy simulation (BES) programs relies heavily on the accuracy of occupant schedule inputs, which have historically been modeled using static, deterministic profiles published decades ago.¹ Standardized schedules represent 24-hour time series of fractions of nominal space occupancy, lighting, or equipment loads.¹ However, real building occupancy is highly stochastic and heterogeneous in both space and time.⁴ Using static profiles yields identical load shapes across spaces, neglecting the spatial-temporal diversity that dictates real building peak loads and heating, ventilation, and air conditioning (HVAC) operations.² While advanced generative models—such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and tabular diffusion models—are increasingly deployed to synthesize realistic schedules⁶, validating these synthetic schedules requires a multi-scale analytical framework. This framework must evaluate distributional realism, sequence-level temporal dynamics, and downstream energy-side consequences to ensure physical and thermodynamic consistency.⁹

A common industry approach validates synthetic diaries using the Jensen-Shannon (JS) divergence per day-type (e.g., separating weekdays and weekends), presence-rate root-mean-squared error (RMS error) in percentage points, and a restricted set of hard rule-based "gates".² While this baseline represents a solid starting point, it is limited in its capacity to verify temporal sequence dynamics, joint probability relationships, and aggregate-versus-individual physical realism.⁵ To address these challenges, this report compiles a comprehensive validation methodology that expands upon this baseline, detailing statistical divergences, sequence preservation metrics, multi-scale checking paradigms, and downstream energy-side consequence testing.

1. Distributional Metrics for Occupancy Realism

Validating synthetic occupancy profiles begins by assessing how closely the statistical distributions of generated schedules match those of observed ground-truth data.⁶ Because human occupancy often transitions between discrete states (e.g., binary presence/absence) or represents continuous fractions of full space capacity⁴, specific statistical distances are employed to quantify discrepancy.

Probability Divergences and Distances

Standard informational divergences are typically deployed to evaluate univariate and joint probability distributions.⁶ Kullback-Leibler (KL) divergence calculates the relative entropy

between the empirical probability distribution of real trajectories $P(x)$ and the generated

distribution $Q(x)$ across a space $\mathcal{X}$ ⁶:

$$KL(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right)$$

While KL divergence is a standard metric for measuring training convergence and distribution matching in Variational Autoencoders (VAEs), it is asymmetric and mathematically undefined if the support of $Q(x)$ does not fully cover $P(x)$ ⁶. The Jensen-Shannon (JS) divergence resolves this asymmetry by computing a symmetric average of KL divergences to a mixture distribution $M = \frac{1}{2}(P + Q)$, bounded between 0 and 1 (using base-2 logarithm) ¹⁵:

$$JS(P \parallel Q) = \frac{1}{2}KL(P \parallel M) + \frac{1}{2}KL(Q \parallel M)$$

For multi-state and multi-modal schedules, the extension to a multi-distribution JS divergence facilitates concurrent evaluation across multiple temporal modalities (e.g., distinguishing weekdays, weekends, and seasonal variations).¹⁶

Despite their mathematical convenience, classic f -divergences like KL and JS suffer from severe limitations when applied to generative models.¹⁵ If the supports of the generated and real distributions are disjoint or have negligible overlap, these divergences saturate, leading to vanishing gradients during adversarial optimization.¹⁵ To overcome this "curse of dimensionality" and geometric insensitivity, researchers utilize the 1-Wasserstein distance, also known as the Earth Mover's Distance (EMD).¹ Representing the optimal transport plan under a cost metric $d(x, y)$, the 1-Wasserstein distance between two distributions μ and ν is defined via Kantorovich-Rubinstein duality as ¹⁵:

$$W_1(\mu, \nu) = \sup_{\|f\|_L \leq 1} (\mathbb{E}_{x \sim \mu}[f(x)] - \mathbb{E}_{y \sim \nu}[f(y)])$$

For one-dimensional cases, such as hourly occupant presence probabilities, this admits a closed-form solution over the cumulative distribution functions $P(u)$ and $Q(u)$ ¹⁸:

$$W_1(\mu, \nu) = \int_{\mathbb{R}} |P(u) - Q(u)| du$$

Computing the exact Wasserstein distance in high dimensions can be computationally expensive, scaling at $\mathcal{O}(n^3)$.¹⁵ Consequently, modern validators leverage entropic regularization variants (e.g., the Sinkhorn divergence) or the sliced Wasserstein distance, which averages one-dimensional projections to achieve a parametric empirical convergence rate of

$n^{-1/2}$ ¹⁵

To evaluate overall sample quality and diversity beyond univariate marginals, the Fréchet Inception Distance (FID) or Maximum Mean Discrepancy (MMD) are applied.⁶ FID compares the mean vector μ and covariance matrix Σ of features extracted from a deep temporal encoder (such as an LSTM or Transformer) for both real (r) and generated (g) populations⁶:

$$FID = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

Temporal and Sequence Realism Metrics

While static probability densities verify whether the overall proportions of occupancy states are maintained, they fail to validate the sequential structure of the time series.¹² High-fidelity synthetic schedules must preserve both short-term temporal dependencies and long-term state persistence.¹² This is validated through two key formulations:

- **Transition Matrix Discrepancy:** Occupancy dynamics can be coarse-grained into transitions between distinct behavioral or presence states.²² Let Z represent the set of discrete states. By modeling run-level transitions where consecutive identical labels are collapsed into distinct runs, the succession of states is captured by a first-order Markov chain transition matrix $P \in |Z| \times |Z|$, estimated via maximum likelihood²²:

$$\hat{P}_{ij} = \frac{N_{ij}}{\sum_{j' \in Z} N_{ij'}}$$

Because self-transitions are omitted at the run level ($\hat{P}_{ii} = 0$), state persistence is decoupled and modeled independently.²² The discrepancy between the real transition matrix $\hat{P}$ and synthetic matrix $\hat{P}_{\text{syn}}$ is computed using the Frobenius norm or mean absolute error (MAE).¹

- **Dwell-Time Distribution Matching:** To capture long-range, non-Markovian temporal structures, validators evaluate the empirical state-conditioned dwell-time distribution D_i for each state $i \in Z$.²¹ The mean dwell time $m_i = \mathbb{E}[d \mid s = i]$ directly dictates the induced frame-level occupancy π_i when linked with the stationary distribution of the transition matrix μ ²²:

$$\pi_i = \frac{\mu_i m_i}{\sum_{\ell \in Z} \mu_\ell m_\ell}$$

Synthetic models must replicate these dwell-time distributions, which often exhibit heavy-tailed or exponential decay characteristics.⁶

- **Autocorrelation Function (ACF):** To verify that the generated time series preserves persistent temporal dependencies and circadian rhythms, the ACF of the synthetic

schedules is calculated across multiple hour lags (e.g., 1 to 24 hours).⁹ The validation metric is defined as the MAE between the empirical ACF of real and synthetic sequences.¹²

Diversity and Coverage Metrics for Generative Models

To prevent the common generative failure of mode collapse, modelers deploy diversity metrics.⁸ Diversity can be evaluated by examining the coverage of the generated dataset over the "uncertainty manifold" of the training data.²⁵ A standard method maps real and synthetic

sequences into a lower-dimensional space using Principal Component Analysis (PCA) or ^t

-Distributed Stochastic Neighbor Embedding (^t-SNE), then computes the overlap of their respective convex hulls or the entropy of the generated distribution.¹³ High-quality models integrate specialized loss terms, such as entropy or KL-divergence regularization within a Wasserstein-gradient penalty (WGAN-GP) framework, to actively promote multi-modal coverage and prevent the generator from collapsing onto a narrow set of highly repetitive "typical" profiles.⁸

Metric Type	Target Attribute	Description	Mathematical Limitation
KL / JS Divergence	Static probability densities	Measures relative entropy and similarity between state probability vectors. ⁶	Insensitive to temporal order; saturates if supports do not overlap. ¹⁵
Wasserstein Distance	Profile shape and timing offsets	Quantifies the minimum "work" required to transform the synthetic schedule into the reference. ¹	Computationally heavy ($\mathcal{O}(n^3)$) for high-dimensional samples without regularization. ¹⁵

Transition Matrix MAE	Sequence transition rates	Compares maximum-likelihood probabilities of moving between occupancy states. ¹²	Ignores the duration of state occupancy (dwell-time). ²²
Dwell-Time KS-Test	State persistence	Compares the empirical CDFs of run durations for each discrete state. ¹²	Does not account for the sequence in which different states are visited. ²²
Autocorrelation MAE	Temporal dependency	Compares the correlation structure of the sequence across various hour lags. ⁹	Captures linear dependencies only; misses non-linear temporal dynamics. ¹²
FID / MMD	Feature-space realism and diversity	Measures distance between deep feature representations of the real and synthetic populations. ⁶	Requires training a feature-extraction network; dependent on encoder quality. ⁶

2. Aggregate-versus-Individual Validation

A major obstacle in validating synthetic building occupancy schedules is the mathematical trade-off between population-level (aggregate) marginal accuracy and sequence-level (individual) realism.⁵

The Masking Phenomenon

Evaluating a generative model purely at the aggregate level introduces a severe masking vulnerability.⁵ For instance, if a model generates synthetic occupancy schedules for a commercial office, the average presence rate computed across 100 simulated offices may perfectly align with the empirical bimodal working-hour curve.⁴ However, inspection of the individual sequences might reveal that each office experiences highly volatile, rapid state-switching (e.g., occupants entering and exiting the space every 15 minutes).¹³ While these erratic transitions average out to a smooth, realistic curve at the population scale, they are physically implausible and distort downstream energy evaluations.⁴ In a BES tool, such high-frequency transitions trigger continuous cycling of local HVAC terminals and lighting

systems, yielding inflated thermal dynamics and energy consumption.⁴

Conversely, focusing exclusively on individual sequence realism can lead to aggregate distortion.⁵ If a generative model synthesizes highly realistic, smooth daily schedules but fails to represent population heterogeneity (e.g., varying arrival/departure times, flexible lunch breaks), integrating these schedules across a multi-zone building or community model will result in highly synchronized, coincident occupancy events.⁴ This synchronization artificially inflates the aggregate peak load, distorting sizing and grid-interaction analyses for district energy systems.⁴

Scale	Goal	Validation Methods	Consequences of Failure
Aggregate Population	Capture diversity, spatial densities, and demographic constraints.	Chi-Square tests, Standardized Root Mean Squared Error (SRMSE), Joint distribution coverage. ¹⁴	Coincident events, exaggerated community peak loads, and poor diversity. ⁴
Individual Sequence	Ensure realistic temporal progression and physical possibility.	Transition matrix Frobenius norm, Dwell-time distributions, Wavelet multi-timescale analysis. ¹²	Artificial HVAC system cycling, high-frequency volatility, and unrealistic user behavior. ⁴

Dual-Scale Validation Protocols

To evaluate both scales without interference, researchers apply multi-scale validation protocols:

1. **Classifier Two-Sample Tests (C2ST):** Rather than relying on simple marginal checks, validators train a binary machine learning classifier (e.g., an XGBoost or Random Forest model) to distinguish between real and synthetic sequences.¹³ If the classifier cannot distinguish the synthetic schedules from the reference data (achieving an accuracy close to **0.5**), the synthetic schedules are considered to have successfully captured both the marginal statistics and the temporal structures.¹³
2. **Bayesian Network Trait Matching:** During population synthesis, individual agents must be stochastically generated using models (such as Bayesian Networks or Variational Autoencoders) that preserve complex sociodemographic relationships (e.g., the correlation between household income, vehicle ownership, and building occupancy

patterns).¹⁴ These agents are then mapped to specific spatial coordinates using compatibility optimization algorithms (e.g., contrastive learning) that respect physical building capacities and census-tract marginal totals.³⁰

3. **Simultaneity and Coincidence Factors:** At the aggregate community scale, validators compute the Simultaneity Factor to check whether individual-level stochasticity properly dampens coincident load spikes, mimicking the natural smoothing effect observed in real smart meter data.⁵

3. Downstream and Energy-Side Validation

The ultimate test of a synthetic occupancy schedule is its performance within a physical BES engine, such as EnergyPlus, Modelica, or IDA ICE.¹⁰ Occupants act as thermal disturbances, generating sensible, latent, and radiative heat, while directly influencing lighting, equipment, domestic hot water, and HVAC operations.¹

Downstream Calibration Metrics

To validate the energy-side consequences, parallel simulations are executed under identical weather and thermal envelope parameters, varying only the occupancy schedules.¹ The resulting simulated energy profiles are compared against measured baseline consumption using the following metrics:

- **Energy Use Intensity (EUI):** Total annual or monthly energy consumption per unit floor area ($\text{kWh}/\text{m}^2\text{-yr}$).³⁷ To prevent misinterpretation of efficiency when occupant density varies, validators utilize a Full-Time Equivalent Occupant (FTEO) adjusted EUI ($\text{kWh}/\text{FTEO-yr}$).³⁷ As occupant count increases, traditional EUI increases slightly due to added plug and lighting loads, whereas occupancy-adjusted EUI decreases because fixed baseloads are distributed over a larger population.³⁷
- **Peak Demand and Peak Timing:** The peak electrical or thermal load (kW) and the hour at which it occurs.¹ Because deterministic or poorly calibrated schedules overestimate peak capacity by assuming simultaneous occupancy across all spaces, synthetic schedules must replicate the stochastic, non-coincident peak distribution of real operations.¹
- **Normalized Mean Bias Error (NMBE):** Quantifies the systematic bias between simulated energy consumption (driven by synthetic schedules) and measured baseline consumption³⁶:

$$NMBE = \frac{\sum_{i=1}^n (m_i - s_i)}{(n - p) \times \bar{m}} \times 100\%$$

where m_i is the measured energy consumption, s_i is the simulated consumption, n is the number of data points, p is the number of model parameters (typically set to 0 for

pure forward schedule validation), and $\bar{m}$ is the mean of the measured data.³⁹

- **Coefficient of Variation of the Root Mean Square Error (CV(RMSE)):** Measures the localized variability and hourly shape mismatch between simulated and measured loads, remaining free from cancelation errors³⁹:

$$CV(RMSE) = \frac{\sqrt{\frac{\sum_{i=1}^n (m_i - s_i)^2}{n-p}}}{\bar{m}} \times 100\%$$

- **Goodness-of-Fit (GOF) Index:** A composite metric combining NMBE and CV(RMSE) to evaluate the overall mathematical alignment of the load shape³⁹:

$$GOF = \sqrt{\frac{CV(RMSE)^2 + NMBE^2}{2}}$$

Industry Standards and Acceptable Tolerances

Major industry standards, including ASHRAE Guideline 14, the Federal Energy Management Program (FEMP), and the International Performance Measurement and Verification Protocol (IPMVP), dictate strict thresholds for calibrated models.³³

Standard / Guideline	Monthly NMBE Threshold	Monthly CV(RMSE) Threshold	Hourly NMBE Threshold	Hourly CV(RMSE) Threshold	Minimum R2 Recommendation
ASHRAE Guideline 14	±5%	≤	±10%	≤	≥ ³⁹
FEMP	±5%	≤	±10%	≤	≥ ³⁹
IPMVP	±20%	≤	±10%	≤	≥ ³⁹

The Compensating Error Trap

A common pitfall in BES model calibration is the "compensating error trap," where unvalidated schedule calibrations mask structural errors in the physical model.¹⁰ For instance, a calibration

algorithm may achieve low prediction errors (e.g., an NMBE of **1.39%** for space heating) by tuning envelope properties, infiltration rates, or boiler efficiency curves to offset unrealistic

occupancy schedules.³⁶ If the unvalidated occupancy schedule is later replaced with measured presence data, the simulated heating demand can shift significantly (e.g., an increase of **3.96% to 4.19%**), exposing the underlying physical mismatch in the calibrated model.³⁶

Consequently, whole-building energy metrics must be validated hierarchically, confirming physical component inputs (e.g., envelope tightness, RTU curves) and schedule distributions independently before evaluating overall load agreement.¹⁰

4. Cross-Validation Designs for Generative Models

To ensure that generative models capture generalizable human behavior patterns and do not overfit specific training sequences, researchers deploy temporal and longitudinal cross-validation designs.²⁸

Stratified and Spatial-Temporal Partitioning

- **10-Fold Cross-Validation:** To mitigate the highly variable test error estimates associated with simple holdout validation, the empirical data is partitioned into ten disjoint folds.²⁸ Models are sequentially trained on nine folds and validated on the remaining fold, with performance statistics averaged across all iterations to report balanced accuracy, sensitivity, and specificity.²⁸
- **Hold-Out Future Wave / "True Future Test":** In longitudinal setups, the model is trained on a specific historical time frame (e.g., years 1–3) and evaluated on a completely distinct, unobserved future period (e.g., year 4).⁴³ This is critical to verify if the generative model can adapt to gradual shifts in climate, occupant demographics, and building retrofits.⁴³
- **Bi-Cross-Validation:** Particularly useful for multi-zone and network-based occupancy models, this technique partitions both the spatial dimensions (building zones) and temporal dimensions (days/weeks).⁴² Latent states are inferred from one subset of zones in the validation set, while the scoring function is computed on the complementary subset, resolving circularity issues and naturally balancing goodness-of-fit against model complexity.⁴²

Retrospective Backcasting and the Unoccupied/Occupied Separation

The retrospective backcasting method represents a rigorous approach to validate the physical and thermodynamic consistency of occupancy models.³² This is demonstrated in a validation case study of a secondary school building in Sweden.³² To isolate occupant-driven thermal loads (such as window opening or "airing") from raw envelope losses, the validation process is split into two distinct steps³²:

1. **Unoccupied Baseline Verification:** The building's thermal performance (e.g., space heating consumption) is simulated during a holiday or unoccupied period.³² Because there are no occupants, any discrepancy is purely attributable to physical properties (envelope U-values, infiltration, thermal bridges).³² A highly accurate discrepancy of **3.4%** was achieved under these conditions, establishing a robust physical baseline.³²
2. **Occupied Backcasting Regression:** Once the physical model is calibrated, the synthetic

occupancy schedules are introduced for the occupied academic semesters.³² A regression analysis is performed between daily heating power (Q_{heat}) and outdoor temperature (T_{out}).³² The difference between simulated and measured heat power represents the occupant behavior impact, capturing dynamic actions like window opening (airing) and localized equipment usage.³² For the Swedish school case, backcasting identified an occupant-induced continuous thermal load of ⁴⁶:

$$Q_{\text{occupant_loss}} = 11.46 \text{ kW} - 0.41 \text{ kW}/^{\circ}\text{C} \times T_{\text{out}}$$

This functional relationship shows that occupant-driven thermal losses escalate as the outdoor temperature drops, providing a precise physical target that synthetic schedules must replicate to be considered thermodynamically valid.⁴⁶

5. Common Validation Pitfalls and Mitigation Strategies

Applying advanced machine learning architectures to time-series generation introduces specific mathematical and structural failures that standard validation metrics can easily overlook.¹⁰

Pitfall A: Model Selection via Training Loss

Generative models—especially GANs and standard Autoencoders—are notoriously difficult to monitor via training loss curves.⁸ A decreasing GAN loss does not necessarily correlate with realistic sequence generation.¹³ In many cases, standard GANs optimize marginal distributions at specific epochs but experience sudden, catastrophic mode collapse in temporal dimensions, producing schedules with highly volatile, physically impossible switching behaviors.¹³

- **Mitigation:** Model selection should be decoupled from training loss.⁹ Checkpoints should be evaluated using a multi-metric Pareto frontier that actively checks the 1-Wasserstein distance of marginals, temporal autocorrelation preservation (ACF MAE), and downstream thermal response metrics rather than raw loss.⁹ Furthermore, incorporating physical constraints directly into the loss function (e.g., enforcing an electrical active-power constraint $P \approx V \cdot I$ or thermodynamic continuity) anchors the adversarial optimization in physical reality.¹³

Pitfall B: The Teacher-Forcing Illusion

Sequence models (such as Recurrent Neural Networks, LSTMs, and autoregressive Transformers) are commonly trained using **Teacher Forcing**.⁴⁷ During training, the model receives the ground-truth state $y(t)$ from the historical data as input at step $t + 1$ to

predict $y(t + 2)$, rather than using its own generated output $\hat{y}(t + 1)$.⁴⁷ This accelerates convergence and stabilizes training, but it conceals a fatal flaw known as **Exposure Bias**.⁴⁷ At test time (inference), the model operates in an open-loop mode, feeding its own generated outputs back into itself as inputs.⁴⁷ Because the model was never exposed to its own errors during training, small prediction errors compound rapidly in the conditioning context, leading to rapid divergence, unrealistic state-switching, and the degradation of the sequence structure.⁴⁷

- **Mitigation:** Models must be evaluated strictly in open-loop generation modes over extended horizons.⁴⁷ Computational strategies to mitigate exposure bias during training include **Scheduled Sampling** (gradually decreasing the probability of feeding ground-truth tokens over training epochs) and **Professor Forcing** (using an adversarial domain classifier to align the hidden states of teacher-forced and self-generated sequences).⁴⁸ Alternatively, utilizing conditional tabular diffusion frameworks (e.g., TabDDPM) bypasses autoregressive dependency errors by modeling the multivariate distribution over the entire sequence duration simultaneously.⁷

Pitfall C: Overfitting Marginals while Breaking Transitions

A generative model can easily overfit the marginal presence probabilities of an occupancy schedule while completely breaking sequence transitions.¹² For example, a model might correctly learn that there is a **90%** probability of a building being occupied at 10:00 AM and a **10%** probability at 11:00 PM.⁴ However, to satisfy these marginals, the model may generate individual sequences that exhibit rapid, physically impossible transitions between presence and absence within those hours.³

- **Mitigation:** Validators must enforce physical and structural regularizations.¹³ These include incorporating a semi-Markov dynamic layer that constrains state transitions based on empirical dwell-time distributions²², or integrating a physics-informed loss term directly into the generator's objective function (e.g., penalizing unrealistic frequency components in the power-spectral density of the synthetic schedule).⁹

6. Tiered Validation Checklist

To establish a standard validation process, a tiered checklist is formulated. A synthetic occupancy schedule model must pass each tier sequentially; failures in Tier 1 or Tier 2 invalidate the model before expensive downstream energy simulations are executed.¹⁰

Tier	Validation Level	Specific Metric	Formula / Mathematical Definition	Recommended Pass/Fail Threshold
Tier 1	Distributional	Kullback-Leible	$KL(P \parallel Q) =$	$\leq$ for

	Realism	r (KL) Divergence	⁶	arrival times, $\leq$ for departure times. ⁶
Tier 1	Distributional Realism	1-Wasserstein Distance (EMD)	$W_1(P, Q) =$ ¹⁸	$\leq$ (ensuring close alignment of cumulative presence density). ⁹
Tier 1	Distributional Realism	Presence-Rate RMS Error	$RMSE =$ $\sqrt{\frac{1}{n} \sum_{i=1}^n (p_i - \hat{p}_i)^2}$	$\leq$ per day-type (weekday/wee kend). ²
Tier 2	Structural Dynamics	Transition Matrix Frobenius Norm	$\ \hat{P} - \hat{P}\ _F$ ¹²	$\leq$ (verifying realistic state-switchin g rates). ¹²
Tier 2	Structural Dynamics	Dwell-Time Distribution Match	Two-sample Kolmogorov-S mirnov (KS) statistic on active/inactive durations. ¹²	Fail to reject the null hypothesis of identical distributions ($\alpha =$). ¹²
Tier 2	Structural Dynamics	Autocorrelatio n (ACF) MAE	$\frac{1}{24} \sum_{k=1}^{24} \ AC\ $ ⁹	$\leq$ across all hourly lags (verifying circadian rhythms). ¹²
Tier 3	Downstream Energy	Normalized Mean Bias	$\frac{\sum (m_i - s_i)}{(n-p) \cdot \bar{m}} \times$	Monthly:

	Impact	Error (NMBE)	³⁹	$\pm 5\%$; Hourly: $\pm 10\%$. (ASHRAE Guideline 14 criteria) ³⁹
Tier 3	Downstream Energy Impact	Coefficient of Variation of RMSE (CV(RMSE))	$\frac{\sqrt{\sum(m_i - \bar{s})^2 / (n - 1)}}{\bar{m}}$ ³⁹	Monthly: $\leq$; Hourly: $\leq$. (ASHRAE Guideline 14 criteria) ³⁹
Tier 3	Downstream Energy Impact	Peak Demand and Peak Timing	$\Delta P = \max(s_i)$; $\Delta t = t_{\max(s)} - t_1$ ¹	Peak Magnitude: $\leq$; Peak Timing Shift: $\leq$. ¹
Tier 3	Downstream Energy Impact	FTEO-Adjusted EUI Deviation	$\frac{EUI_{syn} - EUI_{real}}{EUI_{real}}$; (per FTEO-yr) ³⁷	$\leq$ deviation from validated empirical building performance indicators. ²⁶

References

- ¹ LBNL / ETH Zurich. *Reviewing the representativeness of standard occupancy schedules: Mean Absolute Error and Earth Mover's Distance as measures of difference in profile scale and shape*. URL: <https://www.research-collection.ethz.ch/bitstreams/d81246b4-b723-4e02-b2b4-e551931de055/download>
- ³⁷ Pacific Northwest National Laboratory (PNNL-26019). *Examining traditional and Full-Time Equivalent Occupant (FTEO) adjusted EUI metrics in building energy performance*. URL: https://www.pnnl.gov/main/publications/external/technical_reports/PNNL-26019.pdf
- [50] Winter Simulation Conference (WSC12). *Building energy modeling (BEM) validation*

- under realistic conditions: schedule-related model errors. URL: <https://informs-sim.org/wsc12papers/includes/files/inv156.pdf>
- [14] Lawrence Berkeley National Laboratory. *Synthetic occupant populations using Bayesian Networks: evaluating joint distributions and Standardized-Root-Mean-Squared-Error (SRMSE)*. URL: https://escholarship.org/content/qt9ks056b4/qt9ks056b4_noSplash_888dab35c06dc623328e7b371024a643.pdf
 - ¹⁵ Technical Report. *Optimal Transport-based metrics, Wasserstein distance, and Kantorovich-Rubinstein duality in modern generative modeling*. URL: <https://www.techrxiv.org/doi/pdf/10.36227/techrxiv.174421302.26260179>
 - ¹⁶ Neural Information Processing Systems (NeurIPS 2020). *Multimodal Jensen-Shannon Divergence (mmJSD) for joint distribution estimation and generative modeling*. URL: https://proceedings.neurips.cc/paper_files/paper/2020/file/43bb733c1b62a5e374c63cb22fa457b4-Paper.pdf
 - [17] Machine Learning Tutorial. *Wasserstein GANs (WGAN) vs. Jensen-Shannon Divergence optimization*. URL: <https://towardsdatascience.com/demystified-wasserstein-gans-wgan-f835324899f4/>
 - [7] ArXiv Preprint. *Conditional Tabular Diffusion Models (TabDDPM) for generating mixed-type building characteristics and energy modeling inputs*. URL: <https://arxiv.org/html/2511.02930v1>
 - ¹² ArXiv Preprint. *Hybrid Hidden Markov Models (HMM) with Poisson-driven jump duration and quantile discretization for synthetic time series*. URL: <https://arxiv.org/html/2603.10202v1>
 - ²² Sensors Journal. *Run-level semi-Markov ethology models, transition matrices, and frame-level occupancy calibrations*. URL: <https://www.mdpi.com/1424-8220/26/10/2977>
 - [47] Machine Learning Q&A. *Teacher forcing in Recurrent Neural Networks: exposure bias and testing-time generation distortions*. URL: <https://stats.stackexchange.com/questions/259333/is-teacher-forcing-more-accurate-than-using-actual-model-output-or-just-faster>
 - ⁴⁸ Machine Learning Tutorial. *Scheduled sampling and Professor Forcing: addressing training-generation discrepancies in recurrent systems*. URL: <https://machinelearningmastery.com/teacher-forcing-for-recurrent-neural-networks/>
 - [35] Buildings Journal. *Agent-based modeling (ABM) co-simulation with EnergyPlus to quantify occupant behavior impacts on thermal systems*. URL: <https://www.mdpi.com/2075-5309/11/1/32>
 - ³⁹ Calibration Methodology Report. *Statistical indices for model validation: Coefficient of Variation of the Root Mean Square Error (CV(RMSE)), NMBE, and Goodness-of-Fit (GOF)*. URL: <https://dadun.unav.edu/server/api/core/bitstreams/2614ca9c-c61a-4b67-bf93-2b8c1c0ffa3e3/content>
 - ³⁶ Journal of Building Performance. *The compensating error trap: occupant behavior calibration vs. physical thermal envelope assumptions*. URL:

- <https://www.tandfonline.com/doi/full/10.1080/13467581.2026.2631876>
- [41] Renewable and Sustainable Energy Reviews. *Empirical validation of building energy simulation model input parameters in multizone commercial buildings*. URL: <https://ideas.repec.org/a/eee/rensus/v188y2023ics1364032123007475.html>
 - ²⁵ Journal of Mechanical Design. *Uncertainty-aware generative design with Conditional Hierarchical Wasserstein GANs and KL-divergence regularization*. URL: <https://asmedigitalcollection.asme.org/mechanicaldesign/article/doi/10.1115/1.4071593/1232303/CH-WGAN-Uncertainty-Aware-Generative-Design-of-3D>
 - ¹³ Master's Thesis (TU Wien). *Generation of Synthetic Smart Meter Data: A Comparison of GANs, WGAN-GP, and TimeGAN architectures with Physics-Informed loss terms*. URL: <https://repositum.tuwien.at/bitstreams/20.500.12708/228542/1/Caka%20Rigon%20-%20026%20-%20Generation%20of%20Synthetic%20Smart%20Meter%20Data%20A%20Comparison%20of%20GAN...pdf>
 - [19] Proceedings of Machine Learning Research (PMLR). *Gaussian-smoothed p-Wasserstein distances, Sobolev norms, and generative modeling convergence rates*. URL: <https://proceedings.mlr.press/v139/nietert21a/nietert21a.pdf>
 - [9] Journal of Sustainable Built Environment. *Physics-informed synthetic data generation: validating temporal autocorrelation, Wasserstein distance, and structural health*. URL: https://www.jstage.jst.go.jp/article/jsceiiai/7/1/7_87/_html/-char/en
 - ¹⁰ IBPSA Proceedings. *Hierarchical model evaluation and virtual testbeds for systematic BEM calibration diagnostics*. URL: https://publications.ibpsa.org/proceedings/bs/2023/papers/bs2023_1482.pdf
 - [31] Master's Thesis (FAUDCN). *Synthetic demand profile generation using CVAEs and ARIMA: evaluating Simultaneity Factors and temporal structures*. URL: https://dcn.nat.fau.eu/wp-content/uploads/FAUDCNvH_AniruddhaMaiti_masterThesis_jun2026.pdf
 - ⁶ IBPSA Proceedings. *Urban building energy modeling (UBEM) occupancy schedule generation: Comparing VAE and LSTM-GAN approaches using GPS mobility data*. URL: https://publications.ibpsa.org/proceedings/bs/2025/papers/bs2025_1861.pdf
 - ² Lawrence Berkeley National Laboratory (LBNL). *Performance evaluation of an Agent-Based occupancy simulation model: Evaluated using Mean Error, RMSE, and Jensen-Shannon distance*. URL: https://eta-publications.lbl.gov/sites/default/files/t_hong_-report-_performance_evaluation_of_an_agent-based_occupancy_simulation_model.pdf
 - ⁴² BioRxiv Preprint. *Bi-cross-validation for dynamic connectivity models: balancing goodness-of-fit against model complexity*. URL: <https://www.biorxiv.org/content/10.64898/2026.04.02.716067v1.full-text>
 - [51] ASHRAE Transactions. *Inhomogeneous Markov Chain and ANN models for predicting high-density library building occupancy*. URL: <https://centaur.reading.ac.uk/107547/8/AT-23-C080.pdf>
 - ⁵ IEEE Transactions on Smart Grid. *Stochastic occupant behavior simulator integrated with building stock models to simulate community load shapes*. URL:

<https://arxiv.org/pdf/2111.01881>

- ⁴⁶ Energies Journal. *Measured and simulated energy use in school buildings: validating airing, window opening, and occupancy behavior via backcasting*. URL: <https://www.mdpi.com/1996-1073/13/9/2325>

Works cited

1. Data-driven occupancy schedules for commercial buildings - ETH Zurich Research Collection, accessed on June 12, 2026, <https://www.research-collection.ethz.ch/bitstreams/d81246b4-b723-4e02-b2b4-e551931de055/download>
2. Performance Evaluation of an Agent-based Occupancy Simulation ..., accessed on June 12, 2026, https://eta-publications.lbl.gov/sites/default/files/t_hong_-_report_-_performance_evaluation_of_an_agent-based_occupancy_simulation_model.pdf
3. Generating Occupancy Profiles for Building Simulations Using a Hybrid GNN and LSTM Framework - MDPI, accessed on June 12, 2026, <https://www.mdpi.com/1996-1073/16/12/4638>
4. Simulation of Occupancy in Buildings - Energy Technologies Area, accessed on June 12, 2026, https://eta.lbl.gov/sites/all/files/tianzhen_hong_-_simulation_of_occupancy_in_buildings_1.pdf
5. Stochastic simulation of residential building occupant-driven energy use in a bottom-up model of the US housing stock - arXiv, accessed on June 12, 2026, <https://arxiv.org/pdf/2111.01881>
6. A novel approach to generate occupancy schedules based on ..., accessed on June 12, 2026, https://publications.ibpsa.org/proceedings/bs/2025/papers/bs2025_1861.pdf
7. A Conditional Diffusion Model for Building Energy Modeling Workflows - arXiv, accessed on June 12, 2026, <https://arxiv.org/html/2511.02930v1>
8. Synthesizing Building Operation Data with Generative Models: VAEs ..., accessed on June 12, 2026, <https://www.merl.com/publications/docs/TR2023-072.pdf>
9. Physics-Informed Synthetic Data Generation Framework for Scalable Digital Twin Applications in Built Environment Monitoring - J-Stage, accessed on June 12, 2026, https://www.jstage.jst.go.jp/article/jsceii/71/1/7_87/_html/-char/en
10. A Virtual Testbed for Robust and Reproducible Calibration of Building Energy Simulation Models - IBPSA Publications, accessed on June 12, 2026, https://publications.ibpsa.org/proceedings/bs/2023/papers/bs2023_1482.pdf
11. Data-Driven Occupancy Profile Identification and Application to the Ventilation Schedule in a School Building - MDPI, accessed on June 12, 2026, <https://www.mdpi.com/1996-1073/17/13/3080>
12. Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion - arXiv, accessed on June 12, 2026, <https://arxiv.org/html/2603.10202v1>
13. Generation of Synthetic Smart Meter Data: A Comparison of GAN Architectures -

- repositUm, accessed on June 12, 2026,
<https://repositum.tuwien.at/bitstream/20.500.12708/228542/1/Caka%20Rigon%20-%202026%20-%20Generation%20of%20Synthetic%20Smart%20Meter%20Data%20A%20Comparison%20of%20GAN...pdf>
14. Generating Synthetic Occupants for Use in Building Performance Simulation - eScholarship.org, accessed on June 12, 2026,
https://escholarship.org/content/qt9ks056b4/qt9ks056b4_noSplash_888dab35c06dc623328e7b371024a643.pdf
 15. Sinkhorn Divergence, Wasserstein Distance, and Their Roles in Generative Modeling: A Comprehensive Review of Optimal Transport a - TechRxiv, accessed on June 12, 2026,
<https://www.techrxiv.org/doi/pdf/10.36227/techrxiv.174421302.26260179>
 16. Multimodal Generative Learning Utilizing Jensen-Shannon-Divergence, accessed on June 12, 2026,
https://proceedings.neurips.cc/paper_files/paper/2020/file/43bb733c1b62a5e374c63cb22fa457b4-Paper.pdf
 17. Demystified: Wasserstein GANs (WGAN) - Towards Data Science, accessed on June 12, 2026,
<https://towardsdatascience.com/demystified-wasserstein-gans-wgan-f835324899f4/>
 18. On the Information Processing of One-Dimensional Wasserstein Distances with Finite Samples, accessed on June 12, 2026,
<https://ojs.aaai.org/index.php/AAAI/article/view/39369/43330>
 19. Smooth p-Wasserstein Distance: Structure, Empirical Approximation, and Statistical Applications, accessed on June 12, 2026,
<https://proceedings.mlr.press/v139/nietert21a/nietert21a.pdf>
 20. High-Dimensional Online Change Point Detection with Adaptive Thresholding and Interpretability | OpenReview, accessed on June 12, 2026,
<https://openreview.net/forum?id=ilhWL2WPvS-eld=YQYWP8qPXB>
 21. Using timescale as a state coordinate reveals the metastable geometry of behavior - arXiv, accessed on June 12, 2026, <https://arxiv.org/html/2605.24135>
 22. Virtual Mice, Real Errors: A Sensor-Aware Generative Framework for In Silico Ethology, accessed on June 12, 2026,
<https://www.mdpi.com/1424-8220/26/10/2977>
 23. Ghost Attractors in Spontaneous Brain Activity: Recurrent Excursions Into Functionally-Relevant BOLD Phase-Locking States - Frontiers, accessed on June 12, 2026,
<https://www.frontiersin.org/journals/systems-neuroscience/articles/10.3389/fnsys.2020.00020/full>
 24. Identification and quantification of irreversibility in stochastic systems - RSC Publishing, accessed on June 12, 2026,
<https://pubs.rsc.org/en/content/articlehtml/2026/cp/d5cp04712a>
 25. CH-WGAN: Uncertainty-Aware Generative Design of Three-Dimensional Aperiodic Architected Cellular Materials Via Signed Distance Fields | J. Mech. Des. | ASME Digital Collection, accessed on June 12, 2026,

- <https://asmedigitalcollection.asme.org/mechanicaldesign/article/doi/10.1115/1.4071593/1232303/CH-WGAN-Uncertainty-Aware-Generative-Design-of-3D>
26. Energy Modeling Guidelines - Burlington Electric Department, accessed on June 12, 2026,
<https://www.burlingtonelectric.com/sites/default/files/inline-files/BED-Modeling-Guidelines.pdf>
 27. Producing High-fidelity Synthetic Population Ensembles at Scale - OSTI, accessed on June 12, 2026, <https://www.osti.gov/servlets/purl/3001770>
 28. Full article: Addressing overfitting in classification models for transport mode choice prediction: a practical application in the Aburrá Valley, Colombia - Taylor & Francis, accessed on June 12, 2026,
<https://www.tandfonline.com/doi/full/10.1080/19427867.2024.2422717>
 29. Seasonal Mobility Model Methodology Extended (Places), accessed on June 12, 2026,
<https://documentation.replicahq.com/docs/seasonal-mobility-model-methodology-extended-places>
 30. A Joint Synthetic Housing-Household Inventory - arXiv, accessed on June 12, 2026, <https://arxiv.org/html/2605.17031v1>
 31. Synthetic profile generation for energy system planning using Artificial Intelligence - FAU DCN-AvH, accessed on June 12, 2026,
https://dcn.nat.fau.eu/wp-content/uploads/FAUDCNAvH_AniruddhaMaiti_masterThesis_jun2026.pdf
 32. Measured and Simulated Energy Use in a Secondary School Building in Sweden—A Case Study of Validation, Airing, and Occupancy Behaviour - IDEAS/RePEc, accessed on June 12, 2026,
<https://ideas.repec.org/a/gam/jeners/v13y2020i9p2325-d355044.html>
 33. A systematic approach to manual calibration and validation of building energy simulation, accessed on June 12, 2026,
<https://www.emerald.com/sasbe/article/15/1/203/1259510/A-systematic-approach-to-manual-calibration-and>
 34. Time-Series Generative Networks for Synthesizing Realistic Scenarios in Occupant-Centric Building Simulation - Mitsubishi Electric Research Laboratories, accessed on June 12, 2026,
<https://www.merl.com/publications/docs/TR2025-043.pdf>
 35. Investigating the Impact of Actual and Modeled Occupant Behavior Information Input to Building Performance Simulation - MDPI, accessed on June 12, 2026,
<https://www.mdpi.com/2075-5309/11/1/32>
 36. Full article: Compensating errors in building energy simulation calibration: quantifying data-driven behavioral uncertainty - Taylor & Francis, accessed on June 12, 2026,
<https://www.tandfonline.com/doi/full/10.1080/13467581.2026.2631876>
 37. The Influence of Occupancy on Building Energy Use Intensity and the Utility of an Occupancy-Adjusted Performance Metric - Pacific Northwest National Laboratory, accessed on June 12, 2026,
https://www.pnnl.gov/main/publications/external/technical_reports/PNNL-26019.p

[df](#)

38. Urban residential building stock synthetic datasets for building energy performance analysis - PMC, accessed on June 12, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10910217/>
39. Validation of Calibrated Energy Models: Common Errors - DADUN, accessed on June 12, 2026, <https://dadun.unav.edu/server/api/core/bitstreams/2614ca9c-c61a-4b67-bf93-2b8c1c0ffae3/content>
40. Representing Small Commercial Building Faults in EnergyPlus, Part II: Model Validation, accessed on June 12, 2026, <https://docs.nrel.gov/docs/fy20osti/76026.pdf>
41. Empirical validation of building energy simulation model input parameter for multizone commercial building during the cooling season - IDEAS/RePEc, accessed on June 12, 2026, <https://ideas.repec.org/a/eee/rensus/v188y2023ics1364032123007475.html>
42. Bi-cross-validation: a data-driven method to evaluate dynamic functional connectivity models in fMRI | bioRxiv, accessed on June 12, 2026, <https://www.biorxiv.org/content/10.64898/2026.04.02.716067v1.full-text>
43. PG&E and SDG&E HER Early M&V 2021-2022 Evaluation Plan [2.9.2024 PUBLIC].docx - CPUC, accessed on June 12, 2026, <https://pda.energydataweb.com/api/downloads/3939/PG%26E%20and%20SDG%26E%20HER%20Early%20M%26V%202021-2022%20Evaluation%20Plan%20%5B2.9.2024%20PUBLIC%5D.docx>
44. A data-driven approach for the disaggregation of building-sector heating and cooling loads from hourly utility load data, accessed on June 12, 2026, <https://qsel.columbia.edu/assets/uploads/blog/2023/1-s2.0-S2211467X23001256-main.pdf>
45. Generative AI for Sustainable Smart Environments: A Review of Energy Systems, Buildings, and User-Centric Decision-Making - MDPI, accessed on June 12, 2026, <https://www.mdpi.com/1996-1073/18/23/6163>
46. Measured and Simulated Energy Use in a Secondary School ... - MDPI, accessed on June 12, 2026, <https://www.mdpi.com/1996-1073/13/9/2325>
47. Is teacher forcing more accurate than using actual model output or just faster?, accessed on June 12, 2026, <https://stats.stackexchange.com/questions/259333/is-teacher-forcing-more-accurate-than-using-actual-model-output-or-just-faster>
48. What is Teacher Forcing for Recurrent Neural Networks? - MachineLearningMastery.com, accessed on June 12, 2026, <https://machinelearningmastery.com/teacher-forcing-for-recurrent-neural-networks/>
49. (PDF) A Conditional Diffusion Model for Building Energy Modeling Workflows, accessed on June 12, 2026, https://www.researchgate.net/publication/397321564_A_Conditional_Diffusion_Model_for_Building_Energy_Modeling_Workflows